

Building a .NET Core Application with Middleware

Technology Used: ASP.NET Core

Name: Tushar Dev Singh

Objective: The objective of this assignment is to develop a basic ASP.NET Core web application that demonstrates the use of middleware to handle HTTP requests, serve static files, and implement basic security features.

Features Implemented:

The application includes the following features:

- Middleware for logging requests and responses
- Custom error handling middleware
- Serving static files from wwwroot
- HTTPS enforcement
- Content Security Policy (CSP) security header
- Basic web page using HTML, CSS, and JavaScript

Middleware Implementation

Program.cs:

```
using Microsoft.AspNetCore.Builder;

using Microsoft.AspNetCore.Http;

var builder = WebApplication.CreateBuilder(args);

var app = builder.Build();

// Logging Middleware

app.Use(async (context, next) =>
{
    Console.WriteLine($"Request: {context.Request.Method} {context.Request.Path}");

    await next();

    Console.WriteLine($"Response Status: {context.Response.StatusCode}");

});

// Error Handling Middleware

app.Use(async (context, next) =>
{
    try
    {
        await next();
    }
    catch
    {
        context.Response.ContentType = "text/html";

        await context.Response.WriteAsync("<h1>Custom Error Page</h1>");
    }
});

// Force HTTPS

app.UseHttpsRedirection();

// Content Security Policy Header
```

```

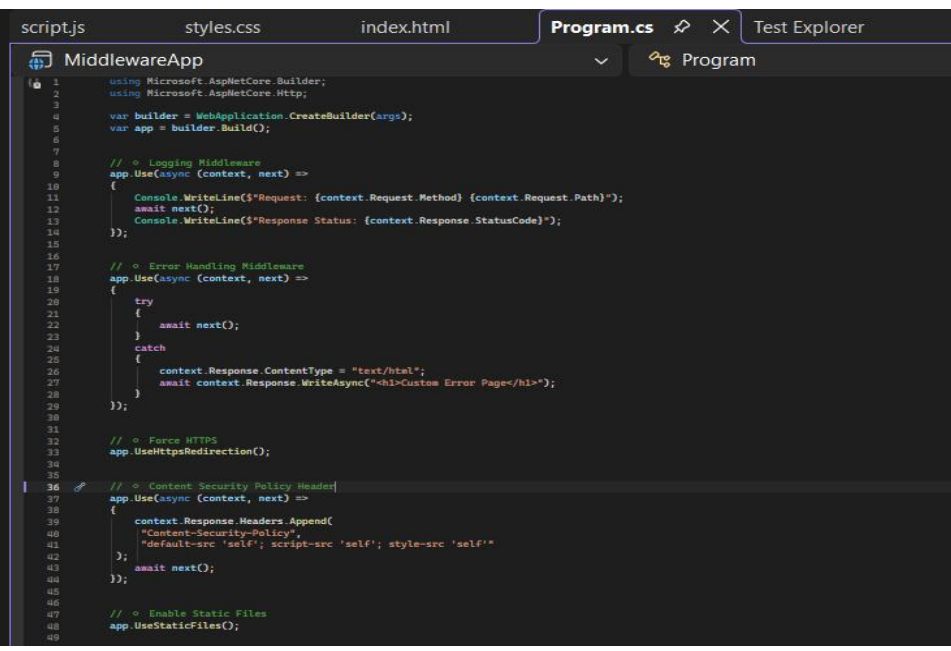
app.Use(async (context, next) =>
{
    context.Response.Headers.Append(
        "Content-Security-Policy",
        "default-src 'self'; script-src 'self'; style-src 'self'"
    );

    await next();
});

// Enable Static Files
app.UseStaticFiles();

// Default route
app.Run(async context =>
{
    await context.Response.WriteAsync("Middleware App Running...");
});

```



```

script.js  styles.css  index.html  Program.cs  Test Explorer
MiddlewareApp
1  using Microsoft.AspNetCore.Builder;
2  using Microsoft.AspNetCore.Http;
3  var builder = WebApplication.CreateBuilder(args);
4  var app = builder.Build();
5
6
7
8  // o Logging Middleware
9  app.Use(async (context, next) =>
10 {
11     Console.WriteLine($"Request: {context.Request.Method} {context.Request.Path}");
12     await next();
13     Console.WriteLine($"Response Status: {context.Response.StatusCode}");
14 });
15
16
17 // o Error Handling Middleware
18 app.Use(async (context, next) =>
19 {
20     try
21     {
22         await next();
23     }
24     catch
25     {
26         context.Response.ContentType = "text/html";
27         await context.Response.WriteAsync("<h1>Custom Error Page</h1>");
28     }
29 });
30
31
32 // o Force HTTPS
33 app.UseHttpsRedirection();
34
35
36 // o Content Security Policy Header
37 app.Use(async (context, next) =>
38 {
39     context.Response.Headers.Append(
40         "Content-Security-Policy",
41         "default-src 'self'; script-src 'self'; style-src 'self'"
42     );
43     await next();
44 });
45
46
47 // o Enable Static Files
48 app.UseStaticFiles();
49
app.Run();

```

Static Files

index.html: <!DOCTYPE html>

<html>

<head>

<title>Middleware App</title>

<link rel="stylesheet" href="styles.css">

</head>

<body>

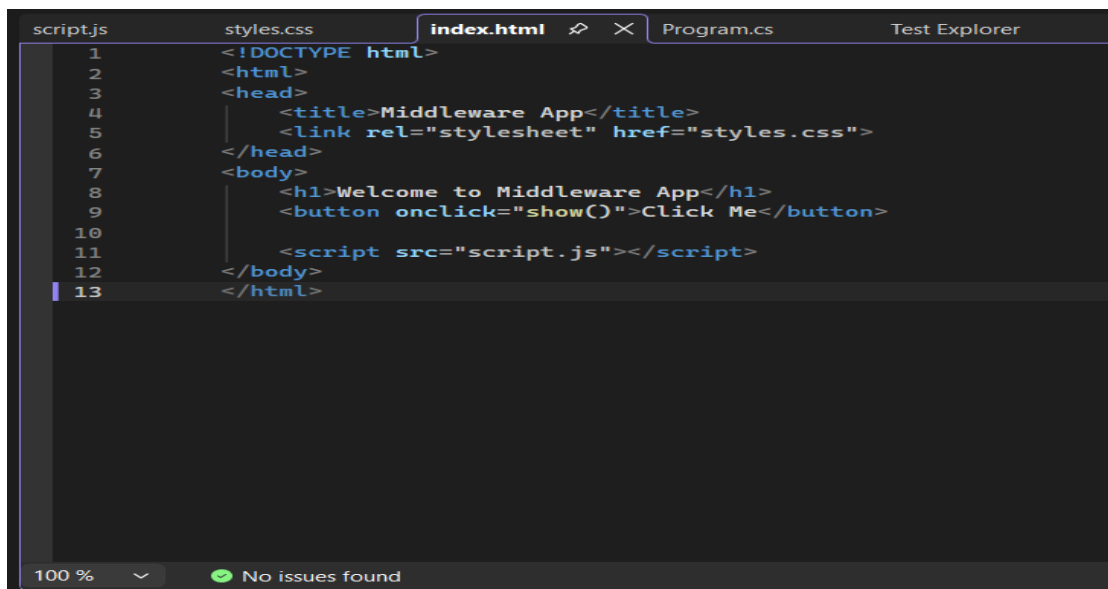
<h1>Welcome to Middleware App</h1>

<button onclick="show()">Click Me</button>

<script src="script.js"></script>

</body>

</html>

A screenshot of a code editor interface. The editor has a dark theme and shows a file named 'index.html' selected in the top tab bar. The code is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Middleware App</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8   <h1>Welcome to Middleware App</h1>
9   <button onclick="show()">Click Me</button>
10
11   <script src="script.js"></script>
12 </body>
13 </html>
```

The left sidebar shows a file explorer with 'script.js', 'styles.css', and 'index.html'. The bottom status bar shows '100 %' and 'No issues found'.

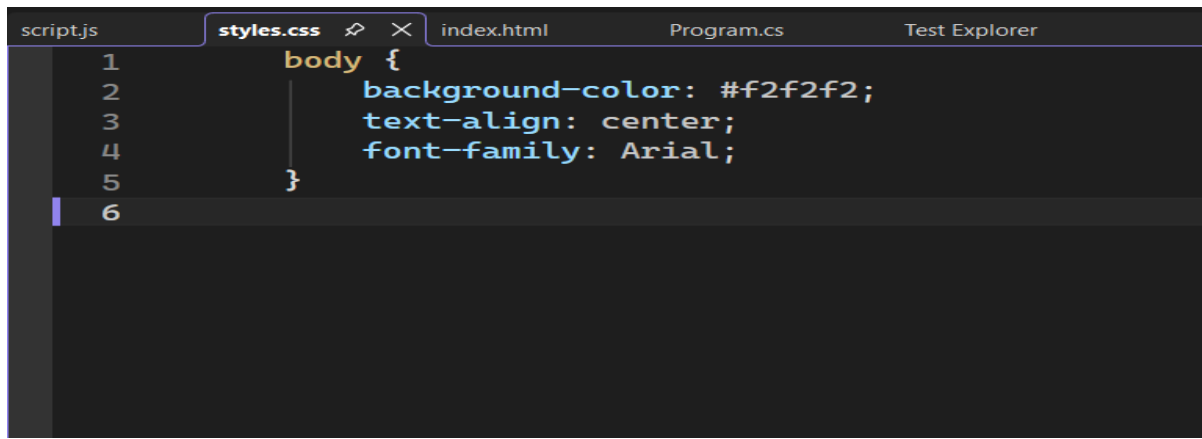
styles.css: body {

background-color: #f2f2f2;

text-align: center;

font-family: Arial;

}

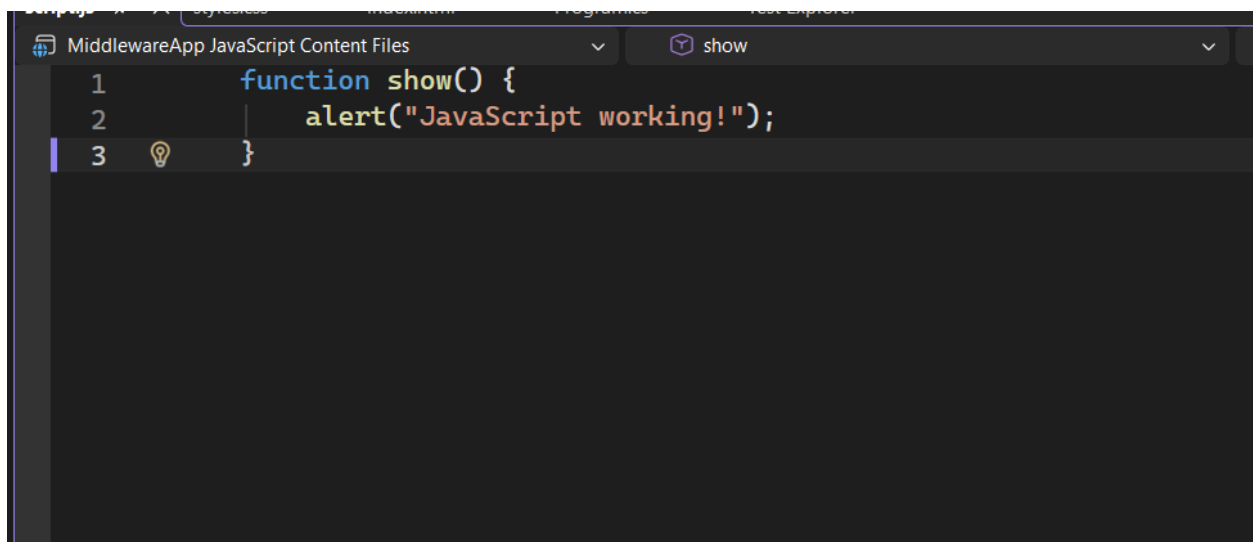
A screenshot of a code editor with a dark theme. The 'styles.css' file is open and selected in the top tab bar. The code is as follows:

```
1      body {
2          background-color: #f2f2f2;
3          text-align: center;
4          font-family: Arial;
5      }
6
```

script.js: function show() {

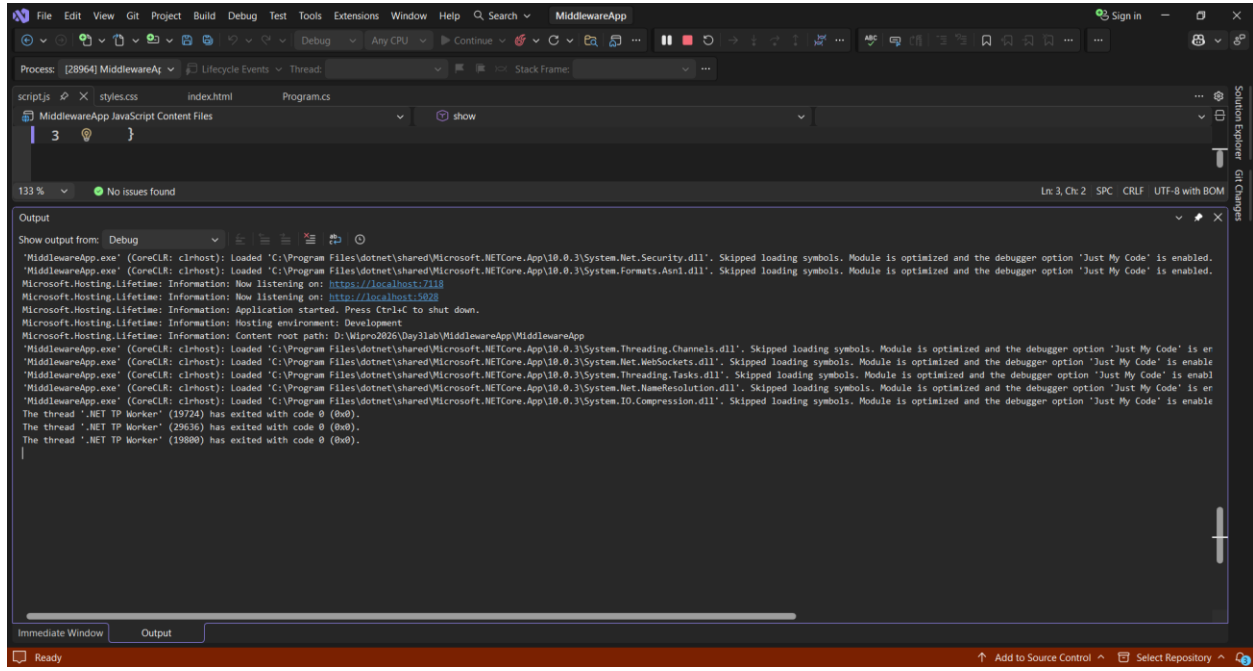
alert("JavaScript working!");

}

A screenshot of a code editor with a dark theme. The 'script.js' file is open and selected in the top tab bar. The code is as follows:

```
1      function show() {
2          alert("JavaScript working!");
3      }
```

Output



The screenshot shows the Visual Studio Code interface with the 'Output' window open. The 'Process' dropdown shows '[28964] MiddlewareApp'. The 'Output' window displays the following log messages:

```
"MiddlewareApp.exe" (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\18.0.3\System.Net.Security.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
"MiddlewareApp.exe" (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\18.0.3\System.Formats.Asn1.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
Microsoft.Hosting.Lifetime: Information: Now listening on: https://localhost:7118
Microsoft.Hosting.Lifetime: Information: Now listening on: http://localhost:5028
Microsoft.Hosting.Lifetime: Information: Application started. Press Ctrl+C to shut down.
Microsoft.Hosting.Lifetime: Information: Hosting environment: Development
Microsoft.Hosting.Lifetime: Information: Content root path: D:\Wipro2026\Day3lab\MiddlewareApp\MiddlewareApp
"MiddlewareApp.exe" (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\18.0.3\System.Threading.Channels.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
"MiddlewareApp.exe" (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\18.0.3\System.Net.Sockets.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
"MiddlewareApp.exe" (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\18.0.3\System.Threading.Tasks.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
"MiddlewareApp.exe" (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\18.0.3\System.Net.NameResolution.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
"MiddlewareApp.exe" (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\18.0.3\System.IO.Compression.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
The thread '.NET TP Worker' (19724) has exited with code 0 (0x0).
The thread '.NET TP Worker' (29636) has exited with code 0 (0x0).
The thread '.NET TP Worker' (19688) has exited with code 0 (0x0).
```

